

CLASS 11TH PHYSICS (THEORY)

ANNUAL BOARD PAPER

Series: 11thARNKD(W/Z) JKLUT-25-1208-C Time: 3.0 Hours

Max. Marks: 70 Marks

Note: Attempt any 60 Marks out of 70 Marks. | Use of standard non-programmable physical constants permitted.

SECTION A: OBJECTIVE TYPE QUESTIONS

1 Mark each

Q1(i). The number of significant figures in 6.032 N m^{-2} is:

- (A) 6 (B) 3 (C) 2 (D) 4

Q1(ii). The correct physical relationship between Calorie and Joule is:

- (A) 1 Calorie = 4.2 Joules (B) 1 Calorie = $\frac{1}{4.2}$ Joules
(C) 0.24 Calorie = 1 Joule (D) Both (a) and (c)

Q1(iii). The horizontal range covered by a projectile fired at angle θ is proportional to:

- (A) its velocity (B) Square of the velocity
(C) Sine of angle of projection (D) Square of sine of angle

Q1(iv). A rocket propulsion system works on the fundamental principle of conservation of:

- (A) Linear momentum (B) Angular momentum (C) Charge (D) Force

Q1(v). The normal circulation of blood through our arteries and veins depends primarily on:

- (A) Surface tension (B) Capillarity (C) Viscosity of blood (D) None of them

Q1(vi). Legible writing on an unglazed newspaper with a fountain pen is difficult because of:

- (A) Inertia (B) Elasticity (C) Capillary action (D) Surface Tension

Q1(vii). The thermodynamic relation governing a reversible adiabatic change of an ideal gas is:

- (A) $P \cdot V = \text{constant}$ (B) $P \cdot V^\gamma = \text{constant}$ (C) $P \cdot V = RT$ (D) None of them

Q1(viii). Two thermodynamic systems are said to be in mutual thermal equilibrium if they have the same:

- (A) Pressure (B) Volume (C) Temperature (D) None of them

Q1(ix). Two vessels A and B of identical volume are at the same temperature. Vessel A holds 1 kg of H_2 gas and vessel B holds 1 kg of N_2 gas. Which vessel contains more molecules?

- (A) A only (B) B only (C) Both contain equal (D) None of them

Q1(x). The expression for the pressure exerted by an ideal gas in terms of gas density (ρ) and rms speed (c) is:

- (A) $\frac{1}{3} \rho c^2 / V$ (B) $\frac{1}{2} \rho c^2 / V$ (C) $\frac{1}{3} \rho c^2$ (D) $\frac{1}{3} \rho c^2 / V$

SECTION B: VERY SHORT ANSWER TYPE QUESTIONS

2 Marks each

Q2. State and establish the law of conservation of angular momentum with a practical illustration.

Q3. Write down the mathematical expressions for angular momentum (L) and work done (W) in rotational motion.

Q4. State any two key physical differences between Mass (inertia of translation) and Moment of Inertia (inertia of rotation).

- Q5.** Obtain the dimensions and SI units of the coefficient of viscosity (η).
- Q6.** What do you mean by a reversible thermodynamic process? Give one representative example.
- Q7.** What are cyclic and non-cyclic thermodynamic processes? Explain briefly.
- Q8.** Write down the four basic postulates / assumptions of the Kinetic Theory of Gases.
- Q9.** What is meant by degrees of freedom of a dynamical system? Write down the degrees of freedom for a rigid diatomic gas molecule.
- Q10.** What is meant by restoring force and spring force constant (k) in an oscillating system?

SECTION C: SHORT ANSWER TYPE QUESTIONS

3 Marks each

- Q11.** Write down three major applications and two limitations of the method of dimensional analysis.
- Q12.** Derive an equation for the trajectory of a projectile fired with velocity u at an angle θ with the horizontal, and prove that the path is a parabola.
- Q13.** A cricket ball is rolled on horizontal ice with an initial velocity of 5.6 m/s and comes to rest after travelling 8 metres. Calculate the coefficient of rolling friction between the ball and ice (take $g = 9.8 \text{ m/s}^2$).
- Q14.** State Newton's First and Second Laws of Motion. Hence define one Newton of force.
- Q15.** A body of mass 10 kg initially at rest is subjected to a constant force of 40 N. Calculate the kinetic energy acquired by the body at the end of 10 seconds.
- Q16.** Distinguish between elastic and inelastic collisions in one dimension with suitable physical examples.
- Q17.** At what height above the earth's surface is the acceleration due to gravity ' g ' exactly the same as in a 100 km deep mine shaft?
- Q18.** Find an expression for the gravitational potential energy of a body of mass m situated at a radial distance r from the centre of the earth.
- Q19.** At what displacement from the mean position in Simple Harmonic Motion (SHM) is the kinetic energy equal to the potential energy? Explain.

SECTION D: LONG ANSWER TYPE QUESTIONS

5 Marks each

- Q20.** What do you mean by resolution of a vector? Find the rectangular horizontal and vertical components of a vector in a Cartesian plane.
- OR ---
- From the velocity-time graph of uniformly accelerated motion, deduce the kinematic equations of motion:
- (i) $v = u + at$
 (ii) $s = ut + \frac{1}{2}at^2$
 (iii) $v^2 - u^2 = 2as$
- Q21.** What do you mean by terminal velocity? Derive an expression for the terminal velocity of a small spherical ball falling through a viscous fluid column using Stokes' Law.
- OR ---
- Explain the three distinct modes of heat transfer: Conduction, Convection, and Radiation with practical daily-life examples.

Q22. Discuss the formation of stationary waves in an open organ pipe and determine the frequencies of normal modes of vibration (harmonics).

--- OR ---

Derive expressions for the instantaneous displacement, velocity, and acceleration of a particle executing simple harmonic motion (SHM).

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